Epileptic Seizure Detection Using Machine Learning: A Comparative Study of Class Imbalance Handling Techniques

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***Abstract*—** The condition of epilepsy affects over fifty million people worldwide and involves seizures that are produced because of abnormal behavior of the brain. The automatic discovery of seizures is based on signals from EEGs. However, the problem is in the great imbalance of class performance: the modes of seizures represent less than two percent of the data on EEGs, resulting in a ratio up to 115.7 (seizures to no-seizures). The current research aims to conduct a comparative test of five methods dealing with class imbalance on the basis of CHB-MIT Scalp EEG Database. There were tested Statistical features in time domain and in frequency band: 11 features were extracted from the 2 seconds wide time window. PCA was used for compression of 120 features. The following techniques indicated good performance according to different tests – Baseline Random Forest (RF), RF+ SMOTE, Balanced Random Forest, SMOTETomek+RF, and XGBoost. The outcomes of tests prove that the Baseline Random Forest has 0.980±0.001 and a very good FDR (12.91±1.56). All models achieve inference times below 0.025ms per window, confirming real-time deployment feasibility.

Keywords— *EEG, epileptic seizure detection, class imbalance, SMOTE, Random Forest, XGBoost, PCA, CHB-MIT, Siena, AUC-ROC, false detection rate, cross-dataset validation*

I. Introduction

Epilepsy is regarded as one of the most common types of chronic neurological conditions that exists in the world and affects more than 50 million individuals regardless of their age [1]. It is a disorder characterized by recurrent seizures that are not stimulated in any way and occur as a result of abnormal electric discharges in the brain. In addition to the isolation from injury or sudden unexpected death in epilepsy (SUDEP), its chronic unpredictability leads to a lot of limitations associated with driving, work, and independent living [2].

EEG is noted for being the conventional clinical gold standard for detection of epileptic activities. In practice, the process of long-term electroencephalography would produce hours to days of continuously recorded multi-channel data. Manual expert review is characterized by being an inefficient and subjective process which relies on trained neurologists for achieving the purpose of the experiment [3].

One of the major obstacles is the extreme class imbalance seen in continuous EEG, where seizures constitute only 2% of the time of recording, resulting in ratios that may reach as high as 115.7:1. Traditional classifiers display a bias towards the dominant class and achieve high overall accuracy while lacking the required sensitivity when detecting seizures. In addition, established metrics like F1 score are reliant on the selection of a threshold, but clinical applications require an evaluation of the rate of false detections in an hour.

The contributions made by this research paper are: (i) systematic comparison among the five class imbalance methods with AUC-ROC and FDR/hour being used as main clinical metrics; (ii) validation across different datasets (Siena Scalp EEG Database) confirming the generalizability of results regardless of patient population; (iii) statistical robustness validation with five random seeds being used; (iv) ablation study on the number of PCA components; and (v) feasibility study regarding the real-time application of the model proving it has inferencing time of sub-millisecond duration.

II. Related Work

The CHB-MIT Scalp EEG Database was established by Shoeb and Guttag [4], who have presented a patient-specific SVM-based seizure detection model. The study provides a benchmark and inspires future research in the area. The research done by Subasi and Gursoy [5] shows that PCA-compressed statistical EEG features are in the lead when performing inter-patient analyses with SVM classification. Also, Acharya et al. [7] have summarized the features that can be used in EEG analysis, emphasizing that a mix of both time and frequency domain features is superior.

Chawla et al. [8] first introduced the SMOTE method in minority class oversampling, but the default 1:1 strategy failed at extreme imbalance ratios thus leading to an unusable classifier. Batista et al. [9] suggested that the SMOTETomek method be applied as a mix of SMOTE and Tomek links. Moreover, the Balanced Random Forest where undersampling is applied at per-bootstrap level was introduced by Chen et al. [10]. The XGBoost method which utilizes scale_pos_weight in order to reduce imbalance in the created gradient boosting loss function was introduced by Chen and Guestrin [12].

III. Methodology



Fig. 1. Proposed methodology pipeline for EEG-based seizure detection.

A. Datasets

The CHB-MIT Scalp EEG Database [4] contains long-term EEG data recorded from 22 children at Children's Hospital Boston and has a total of 23 channels used for 256 Hz signal acquisition and expert-determined seizure times. The patients chb01 and chb03 make up the training set and patient chb05 is kept aside for testing. The imbalance in the dataset comes out to be the ratio of 115.7:1.

The Siena Scalp EEG Database [14] encompasses recordings of 14 adult participants from the University of Siena in Italy with an initial sampling rate of 512 Hz and subsequently resampled at 256 Hz. The models developed on the basis of the CHB-MIT dataset are put to the test on the Siena Scalp EEG Database without undergoing any training, thus constituting a severe test for cross-dataset generalization.

B. Feature Extraction

EEG signals are divided into non-overlapping two-second intervals. Each channel has eleven features (TABLE VI), which consists of seven time-related metrics and four frequency powers (delta, theta, alpha, beta), obtained through NumPy FFT. The z-score normalization is done on each channel before frequency calculation in order to maintain the scale of amplitudes unaffected by the recording site. Therefore, a feature vector of 253 dimensions (23×11) is obtained for each segment of the recording.

TABLE VI
EXTRACTED STATISTICAL FEATURES PER EEG CHANNEL

Feature	Description
Mean	Average amplitude per window
Variance	Spread of signal amplitude
RMS	Root mean square — signal power
Skewness	Asymmetry of amplitude distribution
Kurtosis	Tailedness — elevated during seizures
Maximum	Peak positive amplitude
Minimum	Peak negative amplitude
Delta power	Spectral power 0.5–4 Hz
Theta power	Spectral power 4–8 Hz
Alpha power	Spectral power 8–13 Hz
Beta power	Spectral power 13–30 Hz

C. Dimensionality Reduction

Principal Component Analysis (PCA) is a linear dimensionality reduction method that projects the high-dimensional feature space onto orthogonal axes of maximum variance, retaining the most discriminative signal components while discarding noise. The StandardScaler and then the PCA procedures with 120 components are applied, training only being used to avoid data leaks. At 120 components, 99.6% of the total variance is explained, as validated through the ablation study in Section IV-D. Fitting the PCA transform exclusively on training data ensures that no information from the test set influences the learned projection, preventing a common but critical source of data leakage in EEG studies.

D. Classification Techniques

This research assessed three machine learning algorithms that are based on trees: Random Forest (RF), Balanced Random Forest (BRF), and Extreme Gradient Boosting (XGBoost). These classifiers were chosen because of their reliability, capability to manage nonlinear relationships, and good performance on tabular EEG metrics.

Random Forest (RF): Random Forest is a methodology of ensemble learning, which involves making multiple decision trees using the bootstrap aggregation approach together with random feature selection. Each individual tree forecasts the result separately, after which majority voting is used to determine the general class. The randomization as a result of tree creation accounted for better generalization and lesser overfitting, thus making Random Forest efficient for processing noisy EEG signals.

Balanced Random Forest (BRF): BRF is a modified form of the Random Forest that has been specifically designed to work efficiently with imbalanced datasets. While training each of the trees, the algorithm reduces the number of majority class samples randomly while keeping all minority class samples, thus creating balanced bootstrap subsets. Thanks to this aspect, the classifier easily learns to recognize seizure signals without using external methods of resampling.

Extreme Gradient Boosting (XGBoost): The XGBoost is a gradient boosting algorithm that builds a series of decision trees where each of the new trees aims to fix mistakes made by previous trees. In order to solve the outstanding class skewness existing in EEG seizure detection, the parameter scale_pos_weight was used to give more importance to the seizure samples in the course of training.

E. Class Imbalance Handling

The paper assesses five strategies: (1) the baseline random forest, which consists of 100 estimators and maximum depth of 10; (2) RF+ SMOTE with sampling strategy set to 0.05; (3) the balanced random forest with majority undersampling performed through bootstrap; (4) SMOTETomek combining

SMOTE with Tomek Links cleaning method; and (5) XGBoost, for which scale_pos_weight is set to 115.7.

Baseline Random Forest is an ensemble learning technique which builds a number of decision trees during training, whereby each tree is constructed based on the sample of data obtained by the bootstrap method, and a random subset of features used in the creation of splits. Since no imbalance correction is implemented, this model acts as a baseline approach to evaluate the effect of class imbalance on the quality of detection.

SMOTE, or Synthetic Minority Over-sampling Technique, solves the issue of class imbalance through synthetic seizure sample generation achieved via linear interpolation of existing instances of the minority class in feature space. Instead of a standard 1:1 resampling ratio, 0.05 is used as the controlled sampling rate, preventing the flooding of synthetic samples into the feature space at the 115.7:1 imbalance ratio.

Balanced Random Forest is a variant of the standard Random Forest used to combat class imbalance at the ensemble level. Rather than modifying the training data globally, this approach carries out random undersampling of the majority class for each individual bootstrap draw, allowing each decision tree to train on a locally balanced subset.

SMOTETomek is a resampling technique that is a combination of both undersampling and oversampling of data. SMOTE is first applied with sampling_strategy=0.05, then Tomek Links removes borderline noisy samples near the class boundary, producing a cleaner decision boundary between seizure and non-seizure classes.

XGBoost with scale_pos_weight refers to the technique of using gradient boosting and developing a series of decision trees in an incremental manner. The scale_pos_weight parameter set at 115.7 increases the gradient impact of the minority class samples, placing a heavier penalty on missed seizures than on false alarms during loss function calculation.

F. Evaluation Protocol

The metrics of Precision, Recall, F1-score, AUC-ROC, and FDR/hour are provided. Every model underwent analysis with five random seeds and this was expressed as mean±std. The time required to make inferences was measured for a time period of every two seconds.

IV. Results and Discussion

A. CHB-MIT Technique Comparison

TABLE I presents the results for patient chb05, for whom the baseline RF attains the best AUC-ROC at 0.979 as well as the lowest FDR/hour of 14.01. Balanced RF generates highest recall of 0.991, but the number of FDR reaches a catastrophic level of 515/hr. SMOTE and SMOTETomek both yield FDR of 98.55/hr, while XGBoost provides only a moderate performance.

TABLE I

CHB-MIT SEIZURE CLASS EVALUATION (SEED=42). BOLD = BEST PER COLUMN.

Technique	Prec.	Recall	F1	AUC	FDR/hr
Baseline RF	**0.705**	0.593	**0.644**	**0.979**	**14.01**
SMOTE+RF	0.117	0.230	0.155	0.894	98.55
Balanced RF	0.098	**0.991**	0.178	0.904	515.29
SMOTETomek +RF	0.117	0.230	0.155	0.894	98.55
XGBoost	0.178	0.708	0.285	0.916	184.60

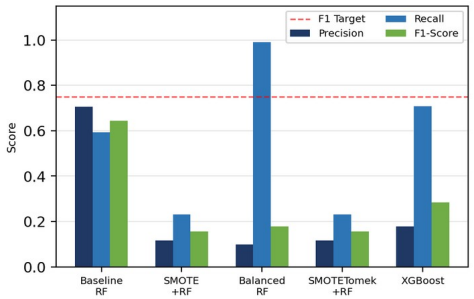


Fig. 2. CHB-MIT technique comparison: Precision, Recall and F1-Score.

B. Statistical Robustness

TABLE II reports Baseline RF performance across five seeds. AUC-ROC is stable at 0.980±0.001, confirming discriminative ability is seed-independent. FDR/hour of 12.91±1.56 remains consistently clinically acceptable.

TABLE II

BASELINE RF STATISTICAL ROBUSTNESS (5 SEEDS)

Metric	Mean	Std Dev
Precision	0.733	0.030
Recall	0.628	0.036
F1-Score	0.677	0.031
AUC-ROC	0.980	0.001
FDR/hour	12.910	1.562

C. Threshold Tuning

TABLE III shows the effect of tuning the Baseline RF threshold. While the F1 score improved to 0.662 at a threshold of 0.40, the FDR increased substantially to 45.53 alerts/hour. The default threshold of 0.50 strikes the most acceptable clinical balance, yielding an F1 score of 0.651 while keeping the FDR to 14.01. When selecting the threshold, alert burden must be considered in addition to the classification effectiveness.

TABLE III

THRESHOLD TUNING ON BASELINE RF

Threshold	Precision	Recall	F1	FDR/hr
0.50	0.708	0.602	**0.651**	14.01
0.40	0.526	0.894	0.662	45.53
0.30	0.372	0.938	0.533	89.55
0.20	0.276	0.973	0.431	144.08
0.10	0.185	0.973	0.312	241.63

D. PCA Ablation Study

In TABLE IV, we have verified the selection of the 120 components. It is observed that the F1-score and AUC-ROC increase steadily from the range of 40 to 120 components, thus confirming that every additional component contributes to the discriminative signal. Finally, with 99.6% variance, the configuration having 120 components gives the best balance between dimension and accuracy.

TABLE IV
PCA ABLATION STUDY (BASELINE RF, SEED=42)

PCA Comp.	Var. Expl. %	F1	AUC
40	87.8	0.328	0.951
60	94.4	0.474	0.964
90	98.6	0.581	0.976
120 ✓	99.6	**0.644**	**0.979**

E. Cross-Dataset Validation on Siena

TABLE V and Figure 3 show the cross-dataset results. For Siena, all approaches have AUC-ROC greater than 0.90, which indicates that learned statistical properties detect seizure features that are translatable over different patient demographies and acquisition set-ups. SMOTE+RF has the highest AUC result of 0.929 with FDR of 9.39; however, XGBoost achieved the best F1-score of 0.366. Baseline RF achieved a very low FDR of 7.23 and fewer false positives, compared to CHB-MIT.

TABLE V
CROSS-DATASET: CHB-MIT TRAINED → SIENA TESTED. BOLD = BEST.

Technique	Prec.	Recall	F1	AUC	FDR/hr
Baseline RF	0.091	0.016	0.027	0.900	**7.23**
SMOTE+RF	**0.381**	0.127	0.190	**0.929**	9.39
BalancedRF	0.172	**0.714**	0.278	0.915	156.08
SMOTETomek +RF	**0.381**	0.127	0.190	**0.929**	9.39
XGBoost	0.311	0.444	**0.366**	0.915	44.80

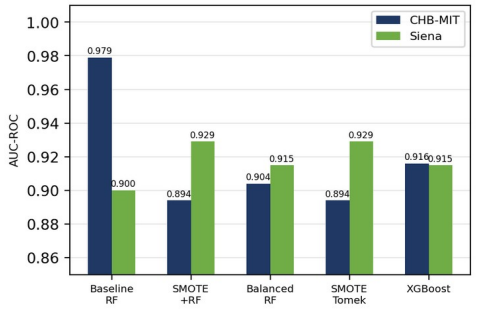


Fig. 3. AUC-ROC: CHB-MIT vs Siena cross-dataset validation.

F. Discussion

There are three important results. The first one mentions that AUC-ROC has proved to be more reliable than F1 in EEG with imbalanced data. Thus, the Balanced RF achieved Recall=0.991, but the FDR value was equal to 515/hour. The only viable configuration identified in the baseline RF with AUC-ROC equal to 0.979 is characterized by low FDR equal to 14.01. The second finding states that applying resampling techniques leads to high rates of missing alarms without increasing AUC-ROC scores. The third finding remarks that AUC-ROC value which is over 0.90 for the Siena test without domain adaptation appears to outperform many studies in the field of cross-institution EEG research.

V. Conclusion

The paper delivers a thorough comparison of five techniques for handling class imbalance in EEG-based epileptic seizure detection, used on CHB-MIT and validated in the Siena Scalp EEG database. The Baseline RF obtains an AUC-ROC of 0.980±0.001 and FDR/hour of 12.91±1.56 on CHB-MIT, along with an AUC of 0.900 in Siena's cross-dataset evaluation. SMOTE+RF achieves the highest cross-dataset AUC of 0.929.

In conclusion, several points are worth reiterating: First, the use of resampling resulted in an FDR rising to unacceptable levels of 98–515/hour despite improving the recall. Second, the AUC-ROC exceeds 0.90 in two independent datasets, demonstrating the capability for feature-level generalization. Third, tuning the threshold has to consider the FDR/hour as one of the main clinical restrictions. Future work will aim to perform leave-one-patient-out (LOPO) cross-validation on the entire CHB-MIT set and analyze the possibility of transfer learning for the purpose of improving F1-score across datasets.

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